

Capstone Section 3 - MySQL

All commands below were run against a local MySQL 8.0.40 server (mysql -u root -p < mysql-queries.sql). Output is captured directly from the server, not simulated. Full script: mysql-queries.sql.

Setup: lms_db database

```
CREATE DATABASE lms_db;
USE lms_db;
```

```
CREATE TABLE users ( user_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100), email VARCHAR(100)
UNIQUE, password VARCHAR(255) );
CREATE TABLE courses ( course_id INT AUTO_INCREMENT PRIMARY KEY, course_name VARCHAR(100),
description TEXT );
CREATE TABLE enrollments ( enrollment_id INT AUTO_INCREMENT PRIMARY KEY, user_id INT, course_id
INT, enrollment_date DATE, FOREIGN KEY (user_id) REFERENCES users(user_id), FOREIGN KEY
(course_id) REFERENCES courses(course_id) );
CREATE TABLE assessments ( assessment_id INT AUTO_INCREMENT PRIMARY KEY, course_id INT, title
VARCHAR(100), max_score INT, FOREIGN KEY (course_id) REFERENCES courses(course_id) );
```

Seeded with the sample users, courses, enrollments and assessments from the assignment brief (Alice Johnson / Bob Smith / Charlie Lee; HTML Basics / CSS Design / MySQL for Beginners, etc).

Task 1: Create instructors table & insert records

```
CREATE TABLE instructors (
    instructor_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);
```

```
INSERT INTO instructors (name, email) VALUES
('Dr. Mei Lin', 'mei.lin@lms.edu'),
('Dr. Arun Kumar', 'arun.kumar@lms.edu'),
('Dr. Wei Zhang', 'wei.zhang@lms.edu');
```

```
SELECT * FROM instructors;
```

Result:

```
+-----+-----+-----+
| instructor_id | name          | email                |
+-----+-----+-----+
| 1             | Dr. Mei Lin   | mei.lin@lms.edu      |
| 2             | Dr. Arun Kumar | arun.kumar@lms.edu  |
| 3             | Dr. Wei Zhang | wei.zhang@lms.edu    |
+-----+-----+-----+
```

Task 2: Enroll Daniel Rose in "CSS Design"

```
INSERT INTO users (name, email, password) VALUES ('Daniel Rose', 'daniel1@lms.com', 'daniel123');

INSERT INTO enrollments (user_id, course_id, enrollment_date)
```

```
VALUES (  
  (SELECT user_id FROM users WHERE email = 'daniel1@lms.com'),  
  (SELECT course_id FROM courses WHERE course_name = 'CSS Design'),  
  CURDATE()  
);
```

```
SELECT u.name, u.email, e.enrollment_date  
FROM enrollments e  
JOIN users u ON e.user_id = u.user_id  
JOIN courses c ON e.course_id = c.course_id  
WHERE c.course_name = 'CSS Design';
```

Result (Bob Smith was already enrolled in CSS Design from the sample data; Daniel Rose is the newly added enrollment, dated today):

+-----+-----+-----+		
name	email	enrollment_date
+-----+-----+-----+		
Bob Smith	bob@example.com	2024-02-15
Daniel Rose	daniel1@lms.com	2026-09-27
+-----+-----+-----+		